Order Selection and Forecasting in Long-Memory Functional Time Series

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ABSTRACT

Functional autoregressive moving-average (FARMA) models describe the temporal dependence of functional time series (FTS), but consistent order selection is challenging under long memory. Within the functional autoregressive fractionally integrated moving-average (FARFIMA) framework, the memory parameter is first estimated, and fractional differencing transforms the long-memory curves into a short-memory process. The FARMA orders are then selected by an **adaptive functional Hannan–Quinn information criterion**, and point forecasts with conformal prediction bands are obtained on the original scale. Simulations compare the criterion with cross-validation and the FARFIMA(1,1,0) benchmark, and applications to fertility and mortality illustrate its practical value.

1 INTRODUCTION

- An FTS is a sequence of curves, such as yearly mortality rates by age.
- FARMA(p, q) models describe their dependence, but **consistent order selection** is challenging.
- Many FTS have **long memory** (Figure 1); common practice simply fixes the memory parameter at $d = 1$.

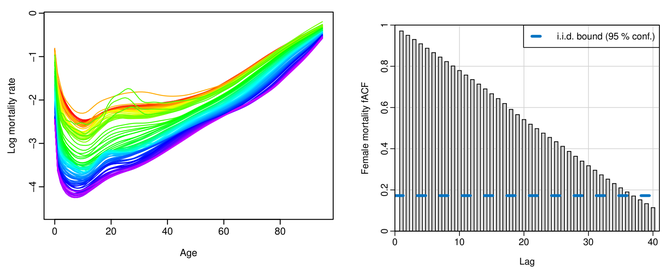


Figure 1. *French female log mortality curves, 1901 (red) to 2023 (violet), and their functional autocorrelation function, which stays above the white-noise bound for over 30 lags.*

RESEARCH QUESTION

How can we select (p, q) consistently, and forecast accurately, under unknown long memory?

CONTRIBUTIONS

- An **adaptive functional Hannan–Quinn criterion** (functional IC), proven consistent.
- A **unified framework** from memory estimation to forecasts with conformal prediction bands.

2 METHODOLOGY

2.1 THE FARFIMA(p, d, q) MODEL

$$(1-B)^d X_t = Y_t,$$

$$Y_t - \sum_{i=1}^{p} \Phi_i Y_{t-i} = \eta_t + \sum_{j=1}^{q} \Theta_j \eta_{t-j}$$

where d is the memory parameter, X_t are the centred curves with **long memory** and Y_t is a **short-memory** FARMA(p, q) process.

- $d = 0$: short memory; $0 < d < ½$: stationary long memory; $d > ½$: non-stationary.

2.2 UNIFIED FRAMEWORK

1 ESTIMATE THE MEMORY

Bias-corrected local polynomial Whittle (BC-LPWN)

$\widehat{d} = \widehat{d}_{\text{BC-LPWN}}$

2 FRACTIONALLY DIFFERENCE

$Y_t = (1-B)^{\widehat{d}} X_t$

3 SELECT THE ORDERS

Functional IC, Section 2.3

$(\hat{p}, \hat{q}) = \arg\min_{p,q \le 5} \mathrm{IC}_T(p, q)$

4 ESTIMATE AND FORECAST

Hannan–Rissanen estimates, plug-in recursion

$\widehat{Y}_{T+h|T} = \sum_{i=1}^{\hat{p}} \widehat{\Phi}_i \widehat{Y}_{T+h-i|T} + \sum_{j=1}^{\hat{q}} \widehat{\Theta}_j \hat{\eta}_{T+h-j|T}$

5 FRACTIONALLY INTEGRATE

Prediction bands, Section 2.4

$\widehat{X}_{T+h|T} = (1-B)^{-\widehat{d}}\, \widehat{Y}_{T+h|T}$

2 METHODOLOGY (CONTINUED)

2.3 ORDER SELECTION BY FUNCTIONAL IC

$$\mathrm{IC}_T(p,q) = \underbrace{\sum_{h=1}^{H_T} \left\| \hat{\Gamma}_h^{(p,q)} \right\|^2}_{\text{fit } F_T(p,q)} + \underbrace{\log\log(T)\, \hat{\delta}_T\, (p+q)}_{\text{adaptive penalty}}$$

with $H_T = \lfloor \log(T) \rfloor$ and noise level $\hat{\delta}_T = \min_{p,q} F_T(p,q)$.

- **Fit:** the remaining dependence of the residuals after fitting FARMA(p, q); near zero only for the true and over-fitted orders.
- **Adaptive penalty:** Hannan–Quinn's $\log\log(T)$ scaled by the noise level; it rules out over-fitting.

Theorem (consistency)

Under our assumptions,

$$\mathbb{P}\{(\hat{p}, \hat{q}) = (p_0, q_0)\} \to 1 \quad \text{as } T \to \infty$$

2.4 CONFORMAL PREDICTION BANDS

Validation errors give a pointwise width, scaled to the nominal coverage by:

Band	Width multiplier
Scaled residual (SR)	calibrated on a grid
Non-conformity score (NS)	split-conformal quantile
Adaptive conformal (AC)	NS level, updated adaptively

3 SIMULATION STUDY

- **Design:** d from 0.2 to 1.4 and true orders (1, 0), (0, 1), (2, 0) or (1, 1), with 100 replications of $T = 1000$ curves.
- **Methods:** functional IC and cross-validation (CV), which minimises the validation root mean squared prediction error (RMSPE).
- **Benchmark:** FARFIMA(1,1,0), which fixes $d = 1$.

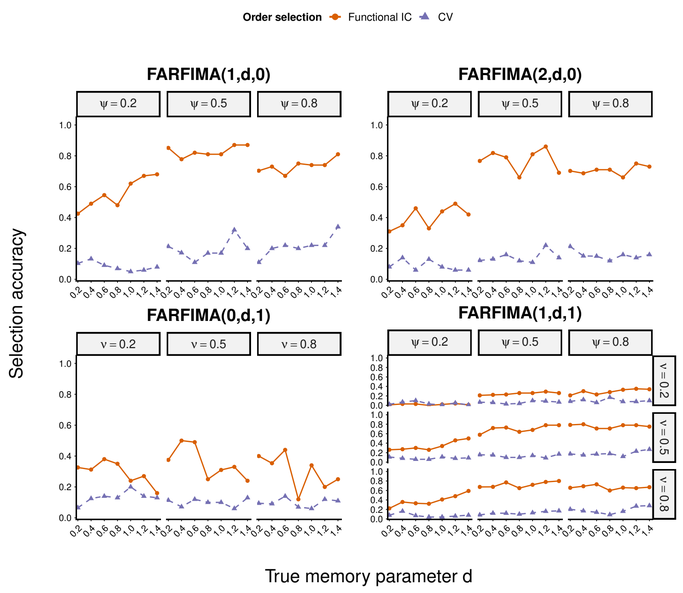


Figure 2. *Share of replications selecting the true orders: functional IC 50% on average, CV 12%.*

Table 1. *Mean RMSPE over all settings.*

Horizon	Benchmark	IC	CV
$h = 1$	0.762	**0.716**	0.718
$h = 5$	2.529	2.383	**2.382**
$h = 10$	4.458	4.231	**4.222**

- Functional IC recovers the true orders **far more often** than CV, with similar forecasting accuracy.
- Gains from d grow with the horizon when $d \neq 1$; when $d = 1$, the orders help at short horizons.

4 REAL DATA APPLICATION

- **Data:** England & Wales fertility and French female and male mortality curves (Table 2).
- **Design:** 60/20/20% training/validation/test split; $h = 1$ to 10; scored on the raw curves.

Table 2. *Data, estimated memory parameter and selected orders; fertility (ages 15–49) is modelled on the Box–Cox scale, mortality (ages 0–95+) on the log scale.*

Series	Years	Est. d	IC	CV
E&W fertility	1938–2022	1.35	(1, 0)	(3, 2)
French female	1901–2023	1.00	(0, 1)	(1, 0)
French male	1901–2023	0.97	(0, 2)	(0, 4)

- Functional IC never selects more short-memory terms than CV.

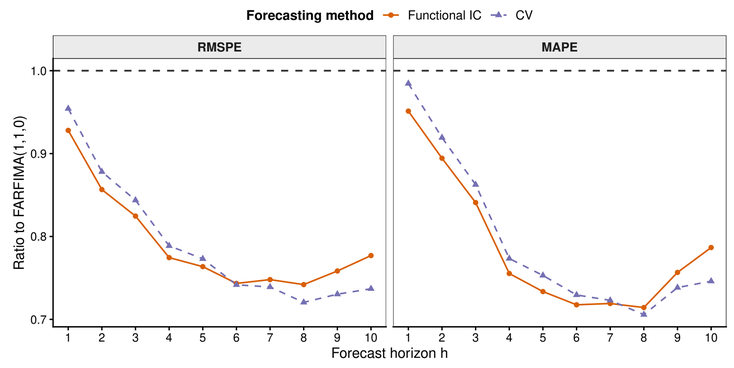


Figure 3. *England & Wales fertility: RMSPE and mean absolute prediction error (MAPE) relative to the benchmark (below 1 is better).*

- Fertility ($d \approx 1.35$): **functional IC cuts the mean RMSPE by 23% and the mean weighted interval score (WIS) by 25%.**

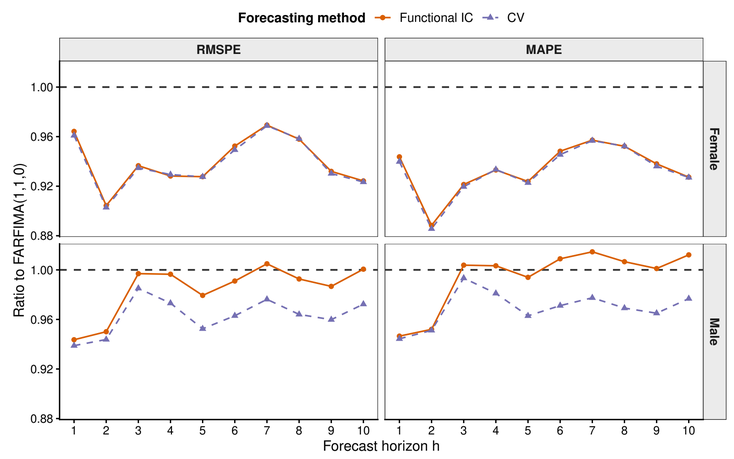


Figure 4. *French female and male mortality: RMSPE and MAPE relative to the benchmark.*

- Mortality ($d \approx 1$): gains come from the orders; both methods cut female RMSPE by about 6%.
- Males: functional IC improves on the benchmark at $h \le 2$; CV improves at every horizon.
- **Bands:** AC brings the empirical coverage closer to the nominal level.

5 CONCLUSIONS

- **Functional IC is recommended:** consistent, parsimonious, and comparable to CV in forecasting accuracy in simulations and real data applications.
- Estimating d matters most when it is far from one; the orders matter most when $d \approx 1$.
- **Future work:** refine d by iterating with the autoregressive strength, and detect change points, such as COVID-19, before estimation.

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